A factorised Pauli–Villars multigrid method for the Möbius domain-wall operator

Working note

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The Möbius domain-wall operator has a spectrum that surrounds the origin, which defeats non-Hermitian Krylov solvers and most multigrid constructions. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator together with the site-diagonal part of the operator, $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$, moves the spectrum into the right half-plane and gives the smallest condition number among the operators they examined. This note describes a two-grid method for A'. The coarse-grid correction is factorised into one Galerkin solve for the quark operator and one for the adjoint Pauli–Villars operator, both with a nearest-neighbour coarse stencil, so that the outer system can be taken on the even sites without enlarging the coarse stencil; the prolongation is built from lifted approximate low modes for which the correction is exact on the vectors it is made from; the smoother is a fixed Chebyshev polynomial. We state the operator and its forms (with Neff's α), the correction, the coarse solvers, the setup, the smoother, the outer solver, and the questions that are open. No numerical results of the method are given: they are still being obtained.

Contents

I Introduction . . . 1
II Spectra of preconditioned Möbius operators . . . 3
III The operator and Pauli–Villars preconditioning . . . 4
IV Even-odd forms . . . 5
V The factorised coarse-grid correction and the prolongation . . . 6
VI The coarse solvers . . . 7
VII Setup . . . 8
VIII The Chebyshev smoother and the cycle . . . 9
IX The outer solver . . . 10
X Open problems . . . 10

I. Introduction

Lattice computations of flavour physics and of the hadronic contributions to precision observables need a fermion discretisation whose chiral symmetry is broken only mildly at finite lattice spacing. The overlap operator [1, 2] has an exact lattice chiral symmetry. Domain-wall fermions [3, 4, 5] approximate it at a lower cost by adding a fifth dimension of extent L_s, in which modes of opposite chirality are bound to the two boundaries by a large negative Wilson mass $-M_5$; the residual chiral symmetry breaking is suppressed as L_s grows, and the effective four-dimensional operator is a rational approximation of the overlap operator. The Möbius generalisation [6, 7, 8] introduces two coefficients (b, c) and improves this approximation at fixed L_s; $c = 0$ is Shamir's operator.

The price is paid in the linear solver. The operator is neither Hermitian nor positive, one application costs L_s four-dimensional Wilson hops, and the standard solver is still the conjugate gradient method on the normal equation of the even-odd preconditioned operator, whose condition number is the square of that of the operator itself. For Wilson-type fermions this problem has been solved by multigrid: the low modes that slow down Krylov solvers are locally coherent [9], so a few of them, cut into blocks, span the low-mode space, and adaptive aggregation multigrid makes the cost nearly independent of the quark mass [10, 11, 12].

Multigrid for domain-wall fermions. The domain-wall operator has a spectrum that encloses the origin, because of the large negative Wilson mass. This violates the half-plane condition of non-Hermitian Krylov methods, whose convergence is then of the order of the system size [13, 14]. Boyle [14] lists four multigrid approaches that have had a measure of success, and they differ in the operator that is coarsened. Cohen *et al.* [15] coarsened the squared unpreconditioned operator, with blocks covering the whole fifth dimension, and noted that its coarse operator reaches two hops. HDCG [16] accelerates the best standard solver, CG on the normal equation of the red-black (even-odd) operator, and coarsens that squared Schur complement; its coarse stencil is next-to-next-to-next-to-nearest neighbour, 81 points for blocks of at least 4^4. HDCR [17] works instead directly on the nearest-neighbour Hermitian indefinite operator $\gamma_5 R_5 D$ at $c = 0$, so that the coarse operator stays nearest neighbour, explicitly to make re-evaluation along an HMC trajectory affordable. Brower, Clark, Howarth and Weinberg [19] pointed out that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$, not D, carries the physical four-dimensional degrees of freedom, solved the system $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ with coarse operators built from the projected Wilson kernel, and demonstrated it in the two-dimensional Schwinger model. Boyle and Yamaguchi [18] compared these approaches for Shamir fermions; HDCR could be set up and solved faster than a single red-black CG solve on large volumes. Two of their observations bear on the present method: among red-black solvers, CG on the Schur complement $M_{ee} - M_{eo} M_{oo}^{-1} M_{oe}$ needed far fewer iterations than on the left-preconditioned $1 - M_{ee}^{-1} M_{eo} M_{oo}^{-1} M_{oe}$; and combining a red-black smoother with the HDCR coarse correction was not effective, which they attributed to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [14] works on the red-black Hermitian operator $(M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})^{\dagger} (M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})$ with a flexible ADEF2 two-level conjugate gradient, a coarse space spanned by blocked low modes obtained with Chebyshev filters, and a stationary coarse solver (deflation plus a Chebyshev polynomial). Treating several right-hand sides at once with block CG and batched matrix products makes the coarse space sub-dominant, and gains of up to a factor of twenty over red-black CG are reported at physical quark mass. In a Lattice 2026 talk, Boyle and Oare [20] named two limitations of this approach—the cost of forming the 81-point coarse operator, and that a single solve on a new gauge field, as in HMC, does not profit from many right-hand sides—and presented a multigrid that directly coarsens the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$ with blocks that contain the whole fifth dimension, which gives a 33-point coarse stencil. They emphasised that $D_{\mathrm{PV}}^{\dagger} D$ is non-normal: its field of values can extend into the left half-plane although its spectrum does not, the Eisenstat–Elman–Schultz bound [21] for GCR is then lost, and they introduced a shift in the smoothers. The coarse solve dominated the cost; a three-level multiple right-hand-side version performs similarly to multiple right-hand-side HDCG, and an exact dense inverse on a further coarse level is planned.

Site-diagonal and Pauli–Villars preconditioning. Kanamori, Chen and Matsufuru [22] examined the spectra of several preconditioned Möbius operators. Combining the Pauli–Villars factor with the site-diagonal part C of the operator, they found that $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$ has eigenvalues with positive real part and the smallest condition number among the candidates, while replacing D_{PV} by a block-diagonal truncation brings back eigenvalues with negative real part. Independently, Neff's modification of the domain-wall matrix [23, 24] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

This note. We describe a two-grid method for A'. The outer system is the usual even-odd reduced one, as in HDCG, but the coarse-grid correction is *factorised*: the inverses of the quark operator and of the adjoint Pauli–Villars operator are replaced by two separate Galerkin inverses, each with a nearest-neighbour coarse stencil and an aggregation that is compatible with the five-dimensional chirality. The prolongation is built from approximate low modes lifted to the whole lattice in two ways, which makes the correction exact on these vectors for the right form of the operator. The smoother is a fixed Chebyshev polynomial

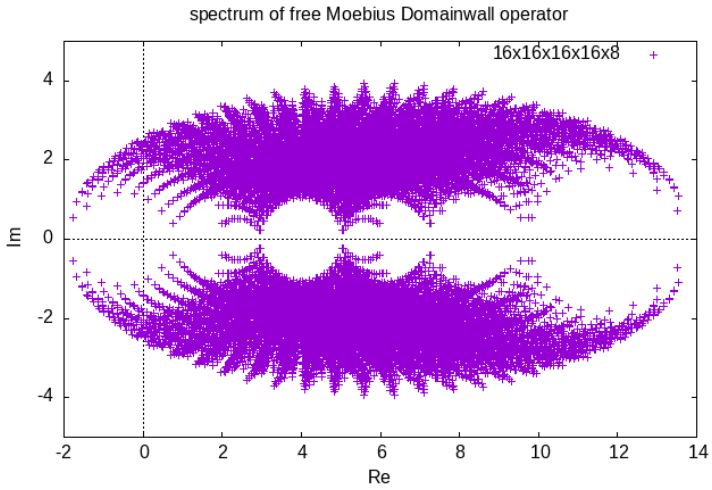


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. Reproduced from Ref. [22], Fig. 1.

and the setup combines Chebyshev filtering with a bootstrap. The note states the method; it does not assess its performance.

Section II recalls the spectra on which the construction rests. Sections III–IX describe the method: the operator and its Pauli–Villars preconditioning, its even-odd forms, the factorised correction and the prolongation, the coarse solvers, the setup, the smoother and the outer solver. Section X lists the open problems. Statements that are our own derivations and have not been checked independently are marked **[derived here; not verified]**.

Conventions. Five-dimensional fields are $\psi_s(x)$, $s = 0, \ldots, L_s - 1$; $P_+, P_- = \frac{1}{2}(1 \pm \gamma_5)$. R reverses the fifth dimension, $(R\psi)_s = \psi_{L_s-1-s}$, and $\Gamma_5 = \gamma_5 R$. Subscripts e and o denote blocks on the even and odd four-dimensional sites; X_{pq} maps parity q to parity p. The injection ι puts a field on the even sites onto the whole lattice (zero on the odd sites), and $\pi = \iota^\dagger$ takes the even sites.

II. Spectra of preconditioned Möbius operators

The location of the spectrum decides which iterative methods can work. A Krylov residual polynomial p with $p(0) = 1$ that is small on the spectrum exists at low degree only if the spectrum can be enclosed by a region, such as an ellipse, that excludes the origin. The same holds for the Chebyshev smoother of Sec. VIII, which is defined by an ellipse in the right half-plane. The spectra of Ref. [22], reproduced in Figs. 1–3, show what the preconditioners do. The lattices are given in the captions.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2 + 1$-flavour configuration at the physical point (Fig. 2, top left) the real parts lie in $[-1.29, 13.2]$. Conjugate gradients and BiCGStab therefore do not apply to D itself, and the practical choice has been CG on a positive definite squared operator.

Site-diagonal preconditioning. With the site-diagonal part C, whose inverse is a constant matrix in the fifth dimension and spin (Sec. B), DC^{-1} has real parts in $[-0.22, 2.22]$ (middle left): the extent shrinks, the sign problem remains.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1} D$, with $D_{\mathrm{PV}} = D(m = 1)$, is a rational approximation of the overlap operator [19]. Because D_{PV}^{-1} is as expensive as a solve with D, Ref. [19] used $D_{\mathrm{PV}}^\dagger$ instead. $D_{\mathrm{PV}}^\dagger D$ has positive real parts (top right), and its low modes lie close to the imaginary axis with a positive

shift of the order of the quark mass, as expected of an approximate overlap operator; its high end, however, reaches about 187, which an overlap operator would not show. Replacing $D_{\rm PV}$ by its block-diagonal approximation $B_{\rm PV}$, which drops the hops between 4^4 blocks, brings back negative real parts (middle and bottom right).

Both together. The operator $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger D C^{-1}$ has real parts in $[2.67\times10^{-4},\ 5.22]$ (Fig. 3, left); its condition number, 1.96×10^4, is smaller than the 2.34×10^4 of $D_{\rm PV}^\dagger D$, and the smallest among the operators examined in Ref. [22]. With $D_{\rm PV}$ replaced by $B_{\rm PV}$ the real parts become negative again (right). A preconditioner that truncates the Pauli–Villars factor to blocks would therefore inherit eigenvalues with negative real part; in the method below the factor is kept exact on the fine lattice and is approximated only through its Galerkin coarse operator (Sec. V).

Two caveats. First, the figures show the full-lattice operator A'; the even-odd operators of Sec. IV are related to it but their spectra are not part of these figures. Second, positivity of the real parts of the eigenvalues is not positivity of the field of values. For a non-normal operator the convergence bound of GCR [21] requires the Hermitian part $(A+A^\dagger)/2$ to be positive definite, which can fail while every eigenvalue lies in the right half-plane; this is the mechanism discussed for $D_{\rm PV}^\dagger D$ in Ref. [20]. Whether it occurs for A' and its even-odd forms is open (Sec. X).

III. The operator and Pauli–Villars preconditioning

A. The Möbius operator

Let K be the hopping term of the Wilson operator,

$$(K\psi)(x) = \sum_{\mu=1}^{4}\Big[(1-\gamma_\mu)U_\mu(x)\psi(x+\hat\mu) + (1+\gamma_\mu)U_\mu^\dagger(x-\hat\mu)\psi(x-\hat\mu)\Big], \tag{3.1}$$

and $W = (4-M_5) - \frac{1}{2}K$ the Wilson operator with mass $-M_5$ and hopping coefficient $\frac{1}{2}$. The hop in the fifth dimension is

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1}, \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1}, \tag{3.2}$$

where m is the quark mass, which enters only through the wrap-around. The Möbius operator is

$$D = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b-cY), \qquad C = d_+ + d_-Y, \tag{3.3}$$

with $d_+ = 1 + b(4-M_5)$ and $d_- = 1 - c(4-M_5)$. K acts on each slice s and does not commute with P_+ and P_-, so the order in Eq. (3.3) matters. The Pauli–Villars operator is the same operator at unit mass, $D_{\rm PV} = D|_{m=1}$, $C_{\rm PV} = C|_{m=1}$.

Γ_5-Hermiticity. Since $\Gamma_5 K \Gamma_5 = K^\dagger$, $\Gamma_5 Y \Gamma_5 = Y^\dagger$ and $\Gamma_5 C \Gamma_5 = C^\dagger$,

$$\Gamma_5 D \Gamma_5 - D^\dagger = -\tfrac{1}{2}\big[K^\dagger,\, b - cY^\dagger\big] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big]. \tag{3.4}$$

The commutator is non-zero because K contains γ_μ and Y contains $P_\pm$. Thus $\Gamma_5 D \Gamma_5 = D^\dagger$ holds for Shamir's operator and fails at $O(c)$ for $c \neq 0$; the Hermitian operator of the Möbius case is $\Gamma_5 D_-^{-1} D$ with $D_- = 1 - cW$ [8], which involves an inverse that is not local. This matters for the coarse solvers (Sec. VI).

B. The site-diagonal part and its inverse

C acts within a four-dimensional site, as an $L_s \times L_s$ matrix in the fifth dimension on each chirality. With $\psi^{\pm} = P_{\pm}\psi$, it is lower bidiagonal on ψ^{+} and upper bidiagonal on ψ^{-}, each with one corner element from the mass:

$$(C\psi)^{+}_{s} = \begin{cases} d_{+}\psi^{+}_{0} + m\, d_{-}\psi^{+}_{L_s-1}, & s = 0, \\ d_{+}\psi^{+}_{s} - d_{-}\psi^{+}_{s-1}, & s \geq 1, \end{cases} \tag{3.5}$$

and the same for ψ^{-} with $s \to L_s - 1 - s$. With $q = d_{-}/d_{+}$, the solution of $C^{+}x = r$ is a forward substitution $p_0 = 0$, $p_s = (r_s + d_{-}p_{s-1})/d_{+}$, followed by

$$x_0 = \frac{r_0 - m\, d_{-}\, p_{L_s-1}}{d_{+} + m\, d_{-}\, q^{L_s-1}}, \qquad x_s = p_s + q^{s}x_0; \tag{3.6}$$

for C^{-} the substitution runs from $s = L_s - 1$ downwards, and for $C^{\dagger}$ the roles of the two chiralities are exchanged. C^{-1} is thus exact and costs $O(L_s)$ per site. It does not depend on the gauge field [22].

C. The preconditioned operator

The operator of Ref. [22] is

$$A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}\, D\, C^{-1}, \qquad A'y = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}\,\eta, \quad \psi = C^{-1}y, \tag{3.7}$$

for $D\psi = \eta$. For $m \to 1$ it becomes the Hermitian positive operator $(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})$.

D. Neff's α

Neff's form of the domain-wall matrix [23, 24] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_{-} + \alpha P_{+}, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_{+} + \alpha P_{-}. \tag{3.8}$$

Q is Hermitian, Q^{-1} is Eq. (3.8) with $\alpha \to 1/\alpha$, and Q leaves $P_{-}\psi_0$ and $P_{+}\psi_{L_s-1}$ unchanged. The site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^{\dagger} = QC^{\dagger}$. The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$. The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D_{\mathrm{PV}}^{(\alpha)} = D_{\mathrm{PV}}Q$ in the latter case.

IV. Even-odd forms

With $D = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}$, $D_{pq} = -\frac{1}{2}K_{pq}(b - cY)$, the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and S_{PV} is that of D_{PV}; note $S^{-1} = \pi D^{-1}\iota$ and $(S_{\mathrm{PV}}^{\dagger})^{-1} = \pi(D_{\mathrm{PV}}^{\dagger})^{-1}\iota$. The site-diagonal factor can stand on either side of S, which gives the right and the left form,

$$\tilde{A}_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^{\dagger}\,(S\,C_e^{-1}), \qquad \tilde{A}_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,(C_e^{-1}S). \tag{4.1}$$

$\tilde{A}_R$ is the even-odd counterpart of A'; $C_e^{-1}S$ is the usual left preconditioned even-odd operator, which Boyle and Yamaguchi found inferior to S itself for CG on the normal equation [18]. With α, DQ has Schur complement SQ_e and diagonal block C_eQ_e, so $\tilde{A}_R$ is independent of α, whereas

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e\,(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,Q_e^{-2}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,Q_e^{-1}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{4.2}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^{\dagger}\eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}C_e^{-1}\eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

TABLE 1. Site-local factors of the correction (5.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1}\iota$; with α the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

V. The factorised coarse-grid correction and the prolongation

A. Why factorise

Solving on the even sites halves the problem and improves the conditioning, but in the literature it has come at a price: the Schur complement is a two-hop operator, so HDCG, which coarsens the squared Schur complement, has an 81-point coarse stencil [16], while the methods with a more compact coarse operator coarsen something other than the Schur complement: $\gamma_5 R_5 D$ (HDCR [17]) or $D_{\mathrm{PV}}^\dagger D$ [19, 20]. The factorised correction avoids this choice: the outer operator is the even-odd one, while the coarse operators are those of D and D_{PV} themselves. By $S^{-1} = \pi D^{-1}\iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e\, S^{-1}\, (S_{\mathrm{PV}}^\dagger)^{-1}\, C_{\mathrm{PV},e}^\dagger = C_e\, \pi D^{-1}\iota\; \pi (D_{\mathrm{PV}}^\dagger)^{-1}\iota\; C_{\mathrm{PV},e}^\dagger, \tag{5.1}$$

a product of two inverses of one-hop operators with site-local factors. The Galerkin operator of $\tilde{A}_R$ itself, a four-hop operator on the fine lattice, would have a much larger coarse stencil; the Galerkin operators of D and D_{PV} are nearest-neighbour, because D is nearest-neighbour in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L\; \pi\mathcal{P}\, \mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger\iota \;\; O_M \;\; \pi\mathcal{P}\left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}\right)^{-1}\mathcal{P}^\dagger\iota \;\; O_R\, r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}, \tag{5.2}$$

with the site-local factors of Table 1. The coarse operators are always those of the plain operators D and D_{PV} (no α), so one setup serves both forms and every α.

B. Aggregation and prolongation

The fine lattice is divided into rectangular four-dimensional blocks; each block contains the whole fifth dimension and becomes one coarse site. Each test vector v is split into its Γ_5 eigencomponents $v_\pm = \frac{1}{2}(1 \pm \Gamma_5)v$, and within each block the $+$ parts and the $-$ parts are orthonormalised separately. With n_v test vectors a coarse site carries $2n_v$ components, the first n_v of positive and the last n_v of negative Γ_5-chirality. Writing $\Sigma = \mathrm{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ on each coarse site,

$$\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma, \tag{5.3}$$

so that $\mathcal{P}^\dagger(\Gamma_5 X \Gamma_5)\mathcal{P} = \Sigma\, \mathcal{P}^\dagger X \mathcal{P}\, \Sigma$ for any fine operator X; in particular $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma$ with $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ whenever $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, i.e. for $c = 0$. Because $v_\pm(L_s - 1 - s) = \pm\gamma_5 v_\pm(s)$, each part is fixed by half of the fifth dimension, so $\mathcal{P}$ need only be stored for $s < L_s/2$. On the even sites the transfer operators are $\mathcal{P}^\dagger\iota$ and $\pi\mathcal{P}$.

C. The basis $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. VII). Each y is lifted in two ways:

$$s = \left(C_e^{-1} y,\; -C_o^{-1} D_{oe} C_e^{-1} y\right), \qquad Ds = \iota\, S C_e^{-1} y, \tag{5.4}$$

$$\hat{t} = \left(a,\; -(C_{\mathrm{PV},o}^\dagger)^{-1}(D_{\mathrm{PV},eo})^\dagger a\right), \quad a = \pi D s, \qquad D_{\mathrm{PV}}^\dagger \hat{t} = \iota\, S_{\mathrm{PV}}^\dagger a, \tag{5.5}$$

where the right-hand columns follow from the block form of D and of $D^\dagger_{\rm PV}$, whose odd-even block is $(D_{{\rm PV},eo})^\dagger$ and whose odd diagonal block is $C^\dagger_{{\rm PV},o}$. The images of s under D and of $\hat{t}$ under $D^\dagger_{\rm PV}$ thus vanish on the odd sites. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

Proposition 1 (Exactness, right form)**.** *Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$, let $\mathcal{D}_{\rm c}$ and $\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P}$ be invertible, and let both coarse systems in Eq.* (5.2) *be solved exactly with the exact adjoint $\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P} = (\mathcal{D}^{\rm PV}_{\rm c})^\dagger$. Then the right-form correction of $r = \tilde{A}_R y$ is $e = y$.*

Proof. For any invertible $\mathcal{P}^\dagger X\mathcal{P}$, $\mathcal{P}(\mathcal{P}^\dagger X\mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$. Now $O_R r = C^\dagger_{{\rm PV},e}\tilde{A}_R y = S^\dagger_{\rm PV} S C_e^{-1} y = S^\dagger_{\rm PV} a$, and $\iota S^\dagger_{\rm PV} a = D^\dagger_{\rm PV}\hat{t}$ by Eq. (5.5). The first coarse solve returns $\pi\hat{t} = a = \pi D s$. With $O_M = 1$ the second has $\mathcal{P}^\dagger\iota\pi D s = \mathcal{P}^\dagger D s$ on the right, since Ds vanishes on the odd sites, and returns $\pi s = C_e^{-1} y$; $O_L = C_e$ gives y. □

Remark 1 (Conditions in practice). **[derived here; not verified]** The proposition needs (i) $s, \hat{t}$ in the range of the stored $\mathcal{P}$, which holds only up to the rounding of the reduced-precision storage of $\mathcal{P}$; (ii) the exact adjoint of the Pauli–Villars coarse factor—the Σ-adjoint $\Sigma\mathcal{D}^{\rm PV}_{\rm c}\Sigma$ of Sec. VI equals it only if $\Gamma_5 D_{\rm PV}\Gamma_5 = D^\dagger_{\rm PV}$, which by Eq. (3.4) fails for $c \neq 0$; (iii) exact coarse solves, whereas they are stopped at a loose tolerance. How much each departure costs has not been analysed.

Remark 2 (Left form). **[derived here; not verified]** For the left form with α in both operators, the same argument requires the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C^{\dagger\,-1}_{{\rm PV},e} Q_e^{-2} C_e^{-1} S Q_e y'$ to lie in $\operatorname{range}\mathcal{P}$ (Table 1 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ one has $s' = s$, but $u = C^{\dagger\,-1}_{{\rm PV},e} C_e^{-1} a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors. We therefore expect the correction not to be exact for the left form with the basis above, but this has not been verified.

D. The coarse operators

$\mathcal{D}_{\rm c} = \mathcal{P}^\dagger D\mathcal{P}$ and $\mathcal{D}^{\rm PV}_{\rm c} = \mathcal{P}^\dagger D_{\rm PV}\mathcal{P}$ are assembled from the hops of the fine operator between neighbouring blocks and from its block-internal part: on each coarse site a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. The adjoint factor $\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P}$ is not assembled; it is applied either exactly, as the conjugate transpose $(\mathcal{D}^{\rm PV}_{\rm c})^\dagger$ of the stored blocks, or approximately as $\Sigma\mathcal{D}^{\rm PV}_{\rm c}\Sigma$. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity and both coarse operators admit an even-odd decomposition of their own.

VI. The coarse solvers

Both coarse systems are solved approximately, to a loose relative residual or a maximum number of iterations, in reduced precision; reaching the tolerance is not required. They are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor.

A. The quark factor

Write $\mathcal{D}_{\rm c} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_{\rm c} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo}A_{oo}^{-1}B_{oe}\right) c_e = g_e - B_{eo}A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe}c_e), \tag{6.1}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. $\mathcal{D}_{\rm c}$ is not Hermitian and $\Sigma\mathcal{D}_{\rm c}$ is Hermitian only for $c = 0$, so a non-Hermitian Krylov method is used.

B. The Pauli–Villars factor

The system is $(\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P})\, w = g$. Two treatments are available.

Exact adjoint. Solve $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^{\dagger} w = g$ with a non-Hermitian Krylov method, preconditioned by the inverses of the adjoint diagonal blocks, $(A_{xx}^{\dagger})^{-1} = (A_{xx}^{-1})^{\dagger}$.

Σ-Hermitian form. Replace the adjoint by $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ and solve the equivalent system

$$\mathcal{H}_{\mathrm{c}}\, w = \Sigma g, \qquad \mathcal{H}_{\mathrm{c}} = \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma, \tag{6.2}$$

with MINRES. $\mathcal{H}_{\mathrm{c}}$ is Hermitian (indefinite) exactly when $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^{\dagger} = \Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$, i.e. for $c = 0$ by Eqs. (3.4) and (5.3); for $c \neq 0$ MINRES is applied to a non-Hermitian matrix and has no guarantee. The preconditioner must be Hermitian positive definite; we use the absolute value of the diagonal blocks [25], $|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^{\dagger}$ with $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^{\dagger}) = V\Lambda V^{\dagger}$ on each coarse site. On a coarse lattice with even extents, the Schur complement of $\mathcal{H}_{\mathrm{c}}$ on the even sites is $S_{\mathrm{c}}\Sigma_e$, with S_{c} the Schur complement of $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$; one solves for $u_e = \Sigma w_e$ on the even sites, completes u on the odd sites as in Eq. (6.1), and sets $w = \Sigma u$.

C. Levels below the coarse lattice

Both factors may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^{\dagger}s_j$ and $\mathcal{P}^{\dagger}\hat{t}_j$, split by Σ and orthonormalised (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbour stencil; the last level is inverted densely. With Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ the diagonal blocks of level ℓ:

Quark factor. A V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1}M_{\ell+1}Q_{\ell+1}^{\dagger}r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{6.3}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system. The level test vectors can optionally be refined by a few GCR steps preconditioned by the levels already built.

Pauli–Villars factor (Σ-Hermitian form only). The additive Hermitian positive preconditioner

$$M_\ell = |B_\ell|^{-1} + Q_{\ell+1}M_{\ell+1}Q_{\ell+1}^{\dagger}, \qquad M_{\ell_{\max}} = |\mathcal{H}_{\ell_{\max}}|^{-1}, \tag{6.4}$$

in the spirit of absolute-value preconditioning [25].

VII. Setup

The setup produces the vectors $y_1, \ldots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$, and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. V).

A. Chebyshev-filtered subspace iteration

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, i.e. eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^{\dagger}\tilde{A}_R$. With the Chebyshev polynomial T_d and $\ell(x) = \big(2x - (\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})\big)/(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $|T_d(\ell(x))| \leq 1$ on $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ while components below λ_{lo} grow; $T_d(\ell(H_n))$ is applied by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

$X \leftarrow n$ random vectors on the even sites

$\lambda_{\rm hi} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\rm lo} \leftarrow$ a small fraction of $\lambda_{\rm hi}$

for each round **do**

$d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, within fixed bounds

$X \leftarrow T_d(\ell(H_n))\, X$; orthonormalise X through its Gram matrix

$G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz

$\lambda_{\rm lo} \leftarrow \theta_{\max}$ ▷ largest Ritz value

end for

return X, Θ

B. Bootstrap

Each bootstrap round builds the basis and the two-grid cycle of Sec. VIII from the current y, replaces every y_j by the result of a fixed small number of FGMRES steps on $\tilde{A}_R x = y_j$ from $x = 0$ preconditioned by that cycle, and orthonormalises the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

$y \leftarrow$ LOWMODES(H_n)

for each bootstrap round **do**

build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it

for $j = 1, \ldots, n$ **do**

$y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$

end for

orthonormalise y

end for

compute $s_j, \hat{t}_j$ by Eqs. (5.4)–(5.5); split by Γ_5; orthonormalise per block $\to \mathcal{P}$

assemble $\mathcal{D}_{\rm c}$, $\mathcal{D}_{\rm c}^{\rm PV}$, the inverses of their diagonal blocks, their even-odd forms

optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

VIII. The Chebyshev smoother and the cycle

A. Smoother

The smoother is ν steps of Chebyshev iteration [26, 27] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\rm lo}, \lambda_{\rm hi}]$: $\lambda_{\rm hi}$ slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\rm lo} = \lambda_{\rm hi}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\rm hi} + \lambda_{\rm lo})$, $\delta = \frac{1}{2}(\lambda_{\rm hi} - \lambda_{\rm lo})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A} d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{8.1}$$

the last step only updating x; ν steps cost $\nu - 1$ applications of $\tilde{A}$. Only a real focal interval is used.

B. Cycle

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (5.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

IX. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as right preconditioner, and ψ is reconstructed as in Sec. IV. Since $\mathcal{M}$ varies from one application to the next, flexible methods are the ones with a theoretical footing: restarted FGMRES and truncated flexible GCR, which keep the preconditioned directions; BiCGStab with the cycle as preconditioner is also available but has no flexible counterpart of its convergence theory. Block variants treat several right-hand sides together. Convergence is monitored on the even-odd system; the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, is evaluated with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator, which is robust but squares the condition number.

X. Open problems

The following are questions of theory and algorithm, not findings.

1. **Exactness for the left form.** Proposition 1 holds for the right form. For $\tilde{A}_L$ (with or without α) the Pauli–Villars lift that would be needed differs from $\hat{t}$ by site-local factors (second remark of Sec. C). Whether a basis can be made exact for both forms at once, and what is lost by using the right-form basis for the left form, is open.

2. **Loss of Γ_5-Hermiticity for $c \neq 0$.** By Eq. (3.4), $\Gamma_5 D \Gamma_5 \neq D^\dagger$ at $O(c)$. MINRES on $\mathcal{H}_c = \mathcal{D}_c^{\mathrm{PV}}\Sigma$, the absolute-value block preconditioner (built from the Hermitian part only), the additive level preconditioner, and the Σ-adjoint in place of the exact adjoint are all exact only for $c = 0$. The exactly Hermitian operator $\Gamma_5 D_-^{-1} D$ [8] involves a non-local inverse. How large the non-Hermitian part is on the coarse space, and how to treat it, is open.

3. **Smoother interval for a non-normal operator.** The interval $[\lambda_{\mathrm{hi}}/\kappa_s, \lambda_{\mathrm{hi}}]$ is set from an estimate of the spectral radius, whereas the setup works with singular values (eigenvalues of $\tilde{A}_R^\dagger \tilde{A}_R$). For a non-normal $\tilde{A}$ these need not agree. Which part of the low end is left to the smoother after the coarse correction, and whether an ellipse with complex foci should replace the real interval, are open.

4. **Even-odd smoother and coarse space.** Boyle and Yamaguchi [18] found that a red-black smoother combined with the HDCR coarse correction was not effective, and attributed it to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned. In the present method the smoother and the correction both approximate $\tilde{A}^{-1}$ and the coarse space is built from low modes of $\tilde{A}_R$, so that

particular mismatch is absent by construction; but the coarse operators are Galerkin projections of D and D_{PV}, not of $\tilde{A}$. Whether the even-odd outer system is preferable to the full-lattice A' for this method has not been established.

5. **Field of values versus real parts of eigenvalues.** Section II shows $\mathrm{Re}\,\lambda > 0$ for A'. The GCR bound [21] needs a positive definite Hermitian part, i.e. a field of values in the right half-plane, which is a stronger condition. Whether it holds for A', $\tilde{A}_R$ and $\tilde{A}_L$, and whether a shift of the kind used in Ref. [20] is needed, is not known.

6. **Outer solver with a varying preconditioner.** Because the coarse solves are truncated, $\mathcal{M}$ is nonlinear and depends on the right-hand side. FGMRES and flexible GCR allow this; BiCGStab does not, and its behaviour with $\mathcal{M}$ as preconditioner has no theoretical justification.

7. **Solving with $D^\dagger$.** The method is built for $D\psi = \eta$. A solve with $D^\dagger$, as needed for the normal equations in HMC, requires the dual correction (the two factors and the roles of the lifts exchanged), and its basis and exactness have not been derived.

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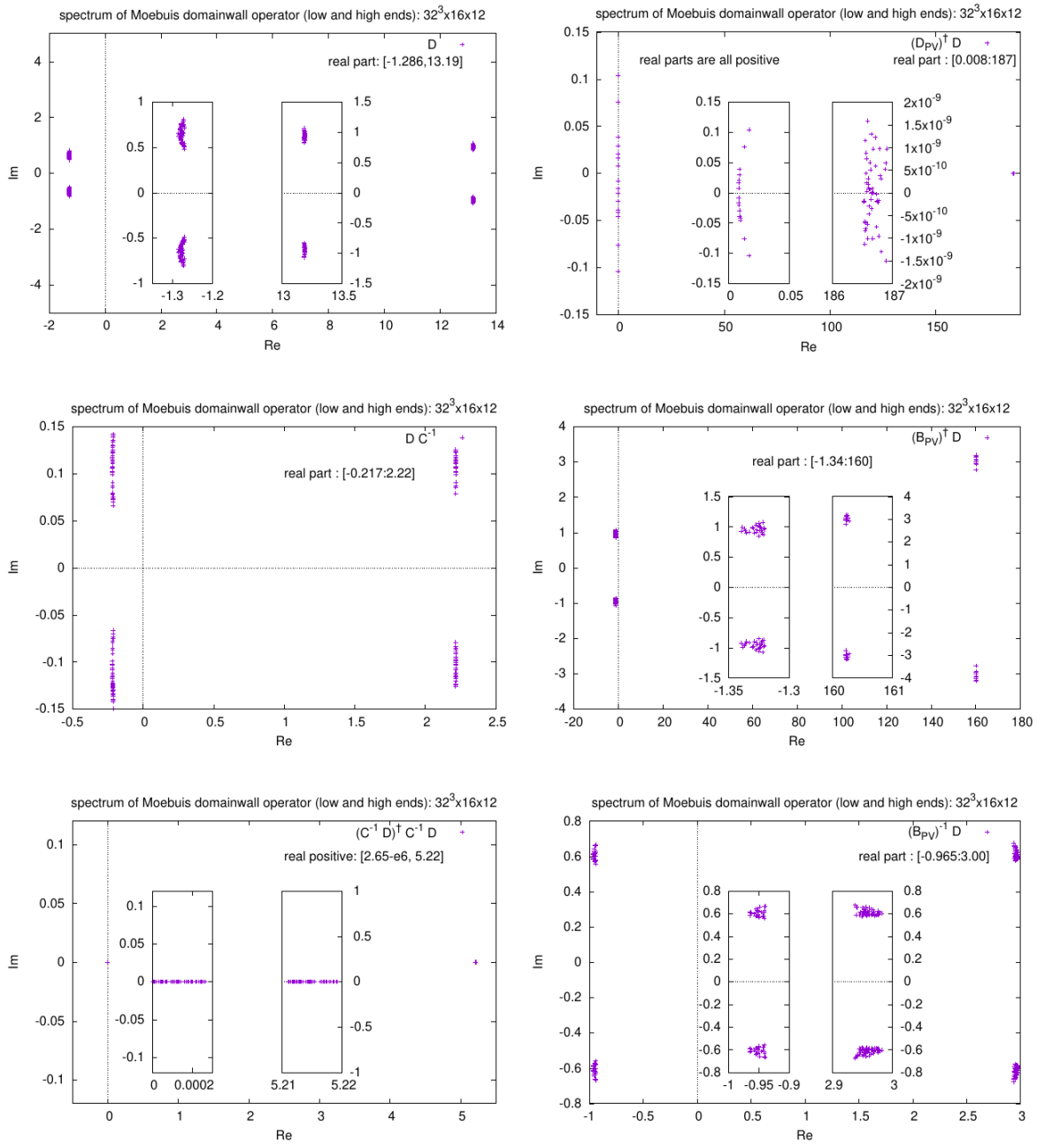


FIG. 2. Low and high ends of the spectrum of the Möbius operator on a JLQCD $2+1$-flavour configuration at the physical point, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D_{\rm PV}^\dagger D$, $B_{\rm PV}^\dagger D$, $B_{\rm PV}^{-1}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). The text of Ref. [22] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. Reproduced from Ref. [22], Fig. 3.

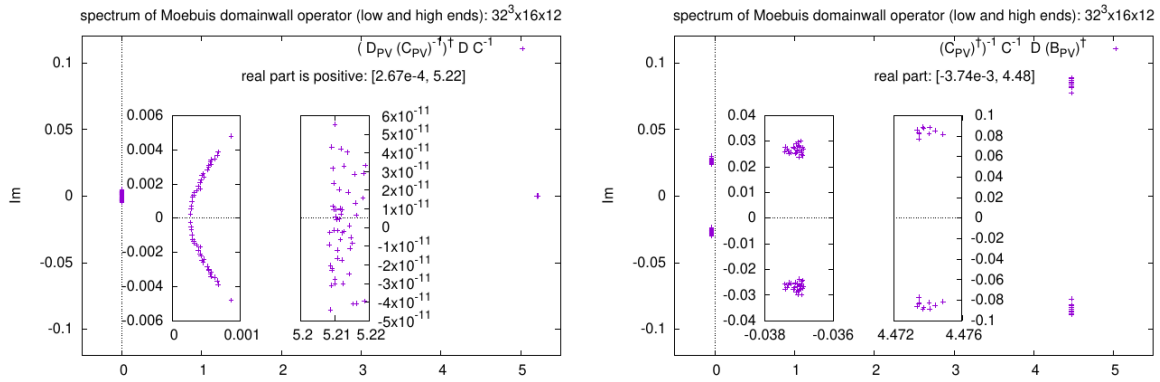


FIG. 3. Left: spectrum of $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger DC^{-1}$, Eq. (3.7), on the configuration of Fig. 2; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. Lattice size as in Fig. 2 (the text of the source quotes $48^3 \times 16$, $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$). Reproduced from Ref. [22], Fig. 4.